

SDS8102: Advanced Mathematical Statistics

Lecture 9: Modes of Convergence and Limit Theory

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Lecture 9: Modes of Convergence and Limit Theory

Suppose $X_1, X_2, \dots$ is a sequence of random variables. What does it mean for

$$X_n \longrightarrow X?$$

There are several answers, depending on *how* we measure the discrepancy between X_n and X .

- Almost sure convergence
- Convergence in probability
- L^p convergence
- Convergence in distribution

These notions are not equivalent:

$$L^p \implies P \implies d, \quad a.s. \implies P.$$

Almost Sure Convergence

Definition:

We say X_n converges **almost surely** to X , written

$$X_n \xrightarrow{a.s.} X,$$

if

$$\mathbb{P} \left(\left\{ \omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \right\} \right) = 1.$$

Thus, outside a set of probability zero,

$$X_1(\omega), X_2(\omega), \dots$$

converges as an ordinary sequence of real numbers.

Equivalently,

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} |X_n - X| = 0 \right) = 1.$$

Example: The Strong Law of Large Numbers states that, under suitable conditions,

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} \mathbb{E}[X_1].$$

Convergence in Probability

Definition:

We say X_n converges **in probability** to X , written

$$X_n \xrightarrow{P} X,$$

if for every $\epsilon > 0$,

$$\mathbb{P}(|X_n - X| > \epsilon) \rightarrow 0.$$

Unlike almost sure convergence, we do not require convergence along almost every sample path.

For each fixed $\epsilon > 0$, we only require the probability of a discrepancy larger than ϵ to vanish.

If

$$\text{Var}(X_n) \rightarrow 0, \quad \mathbb{E}[X_n] \rightarrow \theta \Rightarrow \mathbb{P}(|X_n - \theta| > \epsilon) \leq \frac{\mathbb{E}[(X_n - \theta)^2]}{\epsilon^2} \rightarrow 0,$$

Definition:

For $p \geq 1$, we say

$$X_n \xrightarrow{L^p} X$$

if

$$\mathbb{E}|X_n - X|^p \rightarrow 0.$$

Important special cases:

$$p = 1 : \quad \mathbb{E}|X_n - X| \rightarrow 0,$$

and

$$p = 2 : \quad \mathbb{E}(X_n - X)^2 \rightarrow 0.$$

By Markov's inequality,

$$\mathbb{P}(|X_n - X| > \epsilon) \leq \frac{\mathbb{E}|X_n - X|^p}{\epsilon^p},$$

so

$$X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{P} X.$$

Convergence in Distribution

Let F_n and F denote the CDFs of X_n and X . We say

$$X_n \xrightarrow{d} X$$

if

$$F_n(x) \longrightarrow F(x)$$

at every **continuity point** x of F .

Unlike the previous notions, convergence in distribution only compares the **distributions** of the random variables.

Example. Let

$$X_n \sim N\left(0, 1 + \frac{1}{n}\right).$$

Then

$$F_n(x) = \Phi\left(\frac{x}{\sqrt{1 + 1/n}}\right) \longrightarrow \Phi(x),$$

and hence $X_n \xrightarrow{d} N(0, 1)$.

Importantly, X_n and X need not even be defined on the same probability space.

The Hierarchy of Convergence

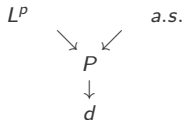
The basic implications are

$$X_n \xrightarrow{a.s.} X \implies X_n \xrightarrow{P} X \implies X_n \xrightarrow{d} X$$

and, for $p \geq 1$,

$$X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{P} X.$$

Thus,



In general, there is no implication between L^p convergence and almost sure convergence.

Most reverse implications are false without additional assumptions.

Almost Sure Convergence Implies Convergence in Probability

Suppose

$$X_n \xrightarrow{a.s.} X.$$

For $\epsilon > 0$, define

$$A_n(\epsilon) = \left\{ \sup_{k \geq n} |X_k - X| > \epsilon \right\}.$$

Then

$$A_1(\epsilon) \supseteq A_2(\epsilon) \supseteq \dots$$

and

$$\bigcap_{n=1}^{\infty} A_n(\epsilon) = \{|X_n - X| > \epsilon \text{ infinitely often}\}.$$

Almost sure convergence gives

$$\mathbb{P} \left(\bigcap_{n=1}^{\infty} A_n(\epsilon) \right) = 0.$$

Almost Sure Convergence Implies Convergence in Probability

Since $A_n(\epsilon) \downarrow$,

$$\mathbb{P}(A_n(\epsilon)) \rightarrow 0.$$

Finally,

$$\{|X_n - X| > \epsilon\} \subseteq A_n(\epsilon),$$

so

$$\mathbb{P}(|X_n - X| > \epsilon) \rightarrow 0.$$

Thus $X_n \xrightarrow{P} X$.

Convergence in Probability Implies Convergence in Distribution

Suppose

$$X_n \xrightarrow{P} X.$$

For any $\epsilon > 0$,

$$\{X_n \leq x\} \subseteq \{X \leq x + \epsilon\} \cup \{|X_n - X| > \epsilon\},$$

so

$$F_n(x) \leq F(x + \epsilon) + \mathbb{P}(|X_n - X| > \epsilon).$$

Similarly,

$$F(x - \epsilon) - \mathbb{P}(|X_n - X| > \epsilon) \leq F_n(x).$$

Therefore,

$$F(x - \epsilon) \leq \liminf_n F_n(x) \leq \limsup_n F_n(x) \leq F(x + \epsilon).$$

If x is a continuity point of F , letting $\epsilon \downarrow 0$ gives

$$F_n(x) \rightarrow F(x).$$

Hence $X_n \xrightarrow{d} X$.

Distribution Does Not Imply Probability

Let

$$X \sim N(0, 1), \quad X_n = -X \quad \text{for every } n.$$

Since X and $-X$ have the same distribution,

$$X_n \stackrel{d}{=} X,$$

and therefore

$$X_n \xrightarrow{d} X.$$

However,

$$|X_n - X| = 2|X|,$$

so for any $\epsilon > 0$,

$$\mathbb{P}(|X_n - X| > \epsilon) = \mathbb{P}\left(|X| > \frac{\epsilon}{2}\right) > 0,$$

independently of n .

Therefore,

$$X_n \xrightarrow{d} X \not\Rightarrow X_n \xrightarrow{P} X.$$

The key issue is that convergence in distribution does not control the **joint coupling** of X_n and X .

An Important Exception: Constant Limits

For any constant c ,

$$X_n \xrightarrow{d} c \iff X_n \xrightarrow{P} c.$$

Only the forward implication needs proof. Suppose

$$X_n \xrightarrow{d} c.$$

For $\epsilon > 0$,

$$\mathbb{P}(|X_n - c| > \epsilon) \leq \mathbb{P}(X_n \leq c - \epsilon) + \mathbb{P}(X_n > c + \epsilon).$$

Since $c - \epsilon$ and $c + \epsilon$ are continuity points of the CDF of the degenerate distribution at c ,

$$F_n(c - \epsilon) \rightarrow 0, \quad F_n(c + \epsilon) \rightarrow 1.$$

Hence

$$\mathbb{P}(|X_n - c| > \epsilon) \rightarrow 0,$$

and therefore

$$X_n \xrightarrow{P} c.$$

Statistical consequence: if $\hat{\theta}_n \xrightarrow{d} \theta$ for a fixed parameter θ , then automatically $\hat{\theta}_n \xrightarrow{P} \theta$.

Probability Does Not Imply Almost Sure Convergence

Convergence in probability allows the “bad event”

$$\{|X_n - X| > \epsilon\}$$

to change with n . Its probability may vanish even if every sample path experiences infinitely many bad events.

Example:

Let $U \sim \text{Unif}(0, 1)$ and define the “typewriter sequence”

$$X_1 = \mathbb{1}_{[0, 1/2)}(U), \quad X_2 = \mathbb{1}_{[1/2, 1)}(U),$$

then

$$X_3 = \mathbb{1}_{[0, 1/4)}(U), \dots, X_6 = \mathbb{1}_{[3/4, 1)}(U),$$

and continue over successive dyadic partitions.

At level k , each interval has length 2^{-k} , $\mathbb{P}(X_n = 1) = 2^{-k} \rightarrow 0 \Rightarrow X_n \xrightarrow{P} 0$.

But for every $U \in (0, 1)$, exactly one interval at each level contains U . Thus $X_n(U) = 1$ infinitely often and $X_n(U) = 0$ infinitely often, so

$$X_n \not\xrightarrow{\text{a.s.}} 0.$$

A Simpler Counterexample via Borel–Cantelli

Let $X_1, X_2, \dots$ be independent with

$$X_n \sim \text{Bernoulli}\left(\frac{1}{n}\right).$$

For every $\epsilon \in (0, 1)$,

$$\mathbb{P}(|X_n| > \epsilon) = \mathbb{P}(X_n = 1) = \frac{1}{n} \longrightarrow 0.$$

Therefore

$$X_n \xrightarrow{P} 0.$$

However,

$$\sum_{n=1}^{\infty} \mathbb{P}(X_n = 1) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

By the second Borel–Cantelli lemma, using independence,

$$\mathbb{P}(X_n = 1 \text{ i.o.}) = 1.$$

Hence X_n equals 1 infinitely often almost surely, and therefore

$X_n \xrightarrow{P} 0$	but	$X_n \not\xrightarrow{\text{a.s.}} 0.$
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Borel–Cantelli and Almost Sure Convergence

Borel–Cantelli gives a useful sufficient condition for upgrading probability bounds to almost sure convergence.

Theorem

If for every $\epsilon > 0$,

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n - X| > \epsilon) < \infty,$$

then

$$X_n \xrightarrow{\text{a.s.}} X.$$

Indeed, the first Borel–Cantelli lemma gives

$$\mathbb{P}(|X_n - X| > \epsilon \text{ i.o.}) = 0.$$

Thus, almost surely, for sufficiently large n ,

$$|X_n - X| \leq \epsilon.$$

Applying this for

$$\epsilon = 1, \frac{1}{2}, \frac{1}{3}, \dots$$

gives

$$X_n \rightarrow X \quad \text{almost surely.}$$

From Probability to Almost Sure Convergence: Subsequences

Theorem

If

$$X_n \xrightarrow{P} X,$$

then there exists a subsequence $\{X_{n_k}\}$ such that

$$\boxed{X_{n_k} \xrightarrow{a.s.} X.}$$

Proof. Since $X_n \xrightarrow{P} X$, for each k choose $n_k > n_{k-1}$ such that

$$\mathbb{P}\left(|X_{n_k} - X| > \frac{1}{k}\right) \leq 2^{-k}.$$

Therefore,

$$\sum_{k=1}^{\infty} \mathbb{P}\left(|X_{n_k} - X| > \frac{1}{k}\right) \leq \sum_{k=1}^{\infty} 2^{-k} < \infty \Rightarrow |X_{n_k} - X| \leq \frac{1}{k}$$

eventually almost surely. Hence $X_{n_k} \xrightarrow{a.s.} X$.

A Useful Characterization of Convergence in Probability

Theorem

The following are equivalent:

$$X_n \xrightarrow{P} X$$

and

every subsequence $\{X_{n_k}\}$ contains a further subsequence $\{X_{n_{k_j}}\}$

such that

$$X_{n_{k_j}} \xrightarrow{\text{a.s.}} X.$$

The forward direction follows by applying the previous theorem to every subsequence, since

$$X_n \xrightarrow{P} X \implies X_{n_k} \xrightarrow{P} X.$$

For the converse, if $X_n \not\xrightarrow{P} X$, then for some $\epsilon, \delta > 0$ there exists a subsequence such that

$$\mathbb{P}(|X_{n_k} - X| > \epsilon) > \delta \quad \text{for all } k.$$

No further subsequence can converge almost surely to X , since almost sure convergence would imply convergence in probability.

L^p Convergence: Relations Between Different p

Suppose $1 \leq p < q$. If

$$X_n \xrightarrow{L^q} X,$$

then

$$X_n \xrightarrow{L^p} X.$$

Indeed, by Lyapunov's inequality,

$$(\mathbb{E}|Y|^p)^{1/p} \leq (\mathbb{E}|Y|^q)^{1/q}.$$

Taking $Y = X_n - X$,

$$\mathbb{E}|X_n - X|^p \leq (\mathbb{E}|X_n - X|^q)^{p/q} \longrightarrow 0.$$

Therefore,

$$L^q \implies L^p \implies P \implies d, \quad q > p \geq 1.$$

The reverse implication $L^p \Rightarrow L^q$ is generally false.

Convergence Does Not Guarantee Convergence of Moments

Suppose

$$X_n \xrightarrow{P} X.$$

Does this imply

$$\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]?$$

In general, **no**.

Example:

Let

$$X_n = \begin{cases} n, & \text{with probability } 1/n, \\ 0, & \text{with probability } 1 - 1/n. \end{cases}$$

Then, for every $\epsilon > 0$,

$$\mathbb{P}(|X_n| > \epsilon) = \frac{1}{n} \rightarrow 0,$$

so

$$X_n \xrightarrow{P} 0.$$

However,

$$\mathbb{E}[X_n] = n \cdot \frac{1}{n} = 1$$

The Problem: Escaping Mass

In the previous example,

$$X_n = n \quad \text{with probability } 1/n.$$

The event $\{X_n \neq 0\}$ becomes increasingly rare:

$$\mathbb{P}(X_n \neq 0) = \frac{1}{n} \rightarrow 0.$$

But its contribution to the expectation remains constant:

$$n \times \frac{1}{n} = 1.$$

More generally,

$$\mathbb{E}|X_n| = \mathbb{E}[|X_n| \mathbb{1}\{|X_n| \leq K\}] + \mathbb{E}[|X_n| \mathbb{1}\{|X_n| > K\}].$$

Convergence in probability controls where most of the probability mass lies, but does not control the second term.

This motivates a condition preventing substantial expectation from hiding in the tails.

Uniform Integrability

Definition:

A collection of integrable random variables $\{X_n\}$ is **uniformly integrable** if

$$\lim_{K \rightarrow \infty} \sup_n \mathbb{E}[|X_n| \mathbb{1}_{\{|X_n| > K\}}] = 0.$$

Compare this with ordinary integrability: for each fixed n ,

$$\mathbb{E}[|X_n| \mathbb{1}_{\{|X_n| > K\}}] \rightarrow 0.$$

Uniform integrability requires this convergence to hold **uniformly over n** .

For

$$X_n = n \mathbb{1}_{\{A_n\}}, \quad \mathbb{P}(A_n) = 1/n,$$

if $n > K$,

$$\mathbb{E}[|X_n| \mathbb{1}_{\{|X_n| > K\}}] = 1.$$

Hence $\{X_n\}$ is not uniformly integrable.

Uniform Integrability Restores Moment Convergence

Uniform integrability is exactly the additional tail control we need.

Theorem

If

$$X_n \xrightarrow{P} X$$

and $\{X_n\}$ is uniformly integrable, then

$$\mathbb{E}|X_n - X| \longrightarrow 0.$$

In particular,

$$\mathbb{E}[X_n] \longrightarrow \mathbb{E}[X].$$

Thus,

$$\text{convergence in probability} + \text{uniform integrability} \implies L^1 \text{ convergence.}$$

This is often referred to as the **Vitali convergence theorem**.

The two assumptions have different roles:

$$X_n \xrightarrow{P} X \quad \text{controls typical behavior,}$$

whereas uniform integrability controls the tails.

A Convenient Sufficient Condition for Uniform Integrability

Checking uniform integrability directly can be difficult.

A useful sufficient condition is a uniform higher-moment bound.

Theorem

If for some $\delta > 0$,

$$\sup_n \mathbb{E}|X_n|^{1+\delta} < \infty,$$

then $\{X_n\}$ is uniformly integrable.

Indeed, on $\{|X_n| > K\}$,

$$|X_n| \leq \frac{|X_n|^{1+\delta}}{K^\delta}.$$

Therefore,

$$\mathbb{E}[|X_n| \mathbb{1}_{\{|X_n| > K\}}] \leq \frac{\mathbb{E}|X_n|^{1+\delta}}{K^\delta}.$$

Taking the supremum over n ,

$$\sup_n \mathbb{E}[|X_n| \mathbb{1}_{\{|X_n| > K\}}] \leq \frac{\sup_n \mathbb{E}|X_n|^{1+\delta}}{K^\delta} \rightarrow 0.$$

Convergence of Higher Moments

Suppose we want

$$\mathbb{E}|X_n|^p \longrightarrow \mathbb{E}|X|^p.$$

Convergence in distribution alone is insufficient.

Theorem

Suppose

$$X_n \xrightarrow{d} X$$

and, for some $q > p$,

$$\sup_n \mathbb{E}|X_n|^q < \infty.$$

Then

$$\boxed{\mathbb{E}|X_n|^p \longrightarrow \mathbb{E}|X|^p.}$$

The higher-moment bound guarantees uniform integrability of $\{|X_n|^p\}$, since

$$\sup_n \mathbb{E}(|X_n|^p)^{q/p} = \sup_n \mathbb{E}|X_n|^q < \infty.$$

Thus weak convergence together with sufficient tail control gives convergence of moments.

Bounded Continuous Functions Behave Better

Although convergence in distribution does not imply convergence of moments, it does imply convergence of expectations for bounded continuous functions.

Theorem

If

$$X_n \xrightarrow{d} X,$$

then for every bounded continuous function f ,

$$\mathbb{E}[f(X_n)] \longrightarrow \mathbb{E}[f(X)].$$

In fact,

$$X_n \xrightarrow{d} X \iff \mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)]$$

for every bounded continuous f .

The Portmanteau Theorem

Convergence in distribution has several equivalent characterizations.

Theorem (Portmanteau)

The following are equivalent:

1. $X_n \xrightarrow{d} X.$

2. *For every bounded continuous f ,*

$$\mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)].$$

3. *For every closed set F ,*

$$\limsup_{n \rightarrow \infty} \mathbb{P}(X_n \in F) \leq \mathbb{P}(X \in F).$$

4. *For every open set G ,*

$$\liminf_{n \rightarrow \infty} \mathbb{P}(X_n \in G) \geq \mathbb{P}(X \in G).$$

These formulations allow us to choose whichever description of weak convergence is most convenient for the problem.

Continuity Sets

Can we conclude

$$\mathbb{P}(X_n \in A) \rightarrow \mathbb{P}(X \in A)$$

for every set A ? **In general, NO.**

A set A is called a **continuity set** of X if

$$\mathbb{P}(X \in \partial A) = 0,$$

where ∂A denotes the boundary of A .

Theorem

If

$$X_n \xrightarrow{d} X$$

and A is a continuity set of X , then

$$\mathbb{P}(X_n \in A) \longrightarrow \mathbb{P}(X \in A).$$

Indeed,

$$A^\circ \subseteq A \subseteq \overline{A},$$

and Portmanteau gives

$$\mathbb{P}(X \in A^\circ) \leq \liminf_n \mathbb{P}(X_n \in A^\circ) \leq \limsup_n \mathbb{P}(X_n \in \overline{A}) \leq \mathbb{P}(X \in \overline{A}).$$

Why Continuity Sets Matter

Consider the deterministic sequence

$$X_n = \frac{1}{n}, \quad X = 0.$$

Clearly,

$$X_n \rightarrow X$$

almost surely, and hence

$$X_n \xrightarrow{d} X.$$

Now take

$$A = \{0\}.$$

For every n ,

$$\mathbb{P}(X_n \in A) = 0,$$

whereas

$$\mathbb{P}(X \in A) = 1.$$

Thus

$$\mathbb{P}(X_n \in A) \not\rightarrow \mathbb{P}(X \in A).$$

The problem is that

$$\partial A = \{0\}$$

and

CDFs as a Convergence-Determining Class

Recall the original definition:

$$X_n \xrightarrow{d} X$$

if

$$F_n(x) \rightarrow F(x)$$

at every continuity point of F .

Why is it enough to check only sets of the form

$$(-\infty, x]?$$

Because

$$F(x) = \mathbb{P}(X \leq x)$$

uniquely determines the entire distribution of X .

Thus the collection

$$\mathcal{C} = \{(-\infty, x] : x \in \mathbb{R}\}$$

is a **convergence-determining class**.

Instead of checking probabilities of every measurable set, it is enough to check a much smaller family that uniquely determines the distribution.

Convergence-Determining Classes

The same principle extends beyond CDFs.

A class of functions $\mathcal{F}$ is convergence-determining if

$$\mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)] \quad \text{for every } f \in \mathcal{F}$$

implies

$$X_n \xrightarrow{d} X.$$

Portmanteau tells us that the class

$$C_b(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{R} : f \text{ bounded and continuous}\}$$

is convergence-determining.

But we can often use much smaller classes.

For example, it is enough to consider bounded Lipschitz functions:

$$|f(x) - f(y)| \leq L|x - y|.$$

The central idea is:

identify a distribution through sufficiently rich test functions.

This perspective will lead naturally to characteristic functions.

The Continuous Mapping Theorem

Once we know $X_n \xrightarrow{d} X$, we often want the limiting distribution of a transformation $g(X_n)$.

Theorem (Continuous Mapping Theorem)

Suppose

$$X_n \xrightarrow{d} X$$

and $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuous. Then

$$g(X_n) \xrightarrow{d} g(X).$$

More generally, it is enough that g be continuous at X almost surely:

$$\mathbb{P}(X \in D_g) = 0,$$

where D_g is the set of discontinuity points of g .

Example. If

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, \sigma^2),$$

then

$$n(\hat{\theta}_n - \theta)^2 \xrightarrow{d} \sigma^2 \chi_1^2.$$

This theorem lets us transform known limiting distributions.

Slutsky's Theorem

Another fundamental tool combines convergence in distribution with convergence in probability.

Theorem (Slutsky)

Suppose

$$X_n \xrightarrow{d} X, \quad Y_n \xrightarrow{P} c,$$

where c is a constant. Then

$$X_n + Y_n \xrightarrow{d} X + c,$$

$$X_n Y_n \xrightarrow{d} cX,$$

and, if $c \neq 0$,

$$\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c}.$$

A common statistical application is replacing an unknown scale parameter by a consistent estimator.

If

$$\frac{\sqrt{n}(\hat{\theta}_n - \theta)}{\sigma} \xrightarrow{d} N(0, 1)$$

and

$$\hat{\sigma}_n \xrightarrow{P} \sigma,$$

Why Slutsky Matters in Statistics

Suppose

$$\frac{\sqrt{n}(\hat{\theta}_n - \theta)}{\sigma} \xrightarrow{d} N(0, 1),$$

but σ is unknown.

If

$$\hat{\sigma}_n \xrightarrow{P} \sigma,$$

then

$$\frac{\sigma}{\hat{\sigma}_n} \xrightarrow{P} 1.$$

Therefore, by Slutsky,

$$\begin{aligned} \frac{\sqrt{n}(\hat{\theta}_n - \theta)}{\hat{\sigma}_n} &= \frac{\sqrt{n}(\hat{\theta}_n - \theta)}{\sigma} \frac{\sigma}{\hat{\sigma}_n} \\ &\xrightarrow{d} N(0, 1). \end{aligned}$$

Thus

unknown population quantities can often be replaced by consistent estimators.

This is the principle behind **studentization** and many asymptotic confidence intervals.

Characteristic Functions

There is another way to characterize a probability distribution.

Definition:

The **characteristic function** of a random variable X is

$$\varphi_X(t) = \mathbb{E}[e^{itX}], \quad t \in \mathbb{R}.$$

Since

$$|e^{itX}| = 1,$$

the characteristic function always exists.

Some basic properties are

$$\varphi_X(0) = 1, \quad |\varphi_X(t)| \leq 1,$$

and

$$\varphi_{aX+b}(t) = e^{itb} \varphi_X(at).$$

Most importantly, the characteristic function uniquely determines the distribution of X .

Characteristic Functions and Independence

Characteristic functions are especially convenient for sums.

If X and Y are independent, then

$$\begin{aligned}\varphi_{X+Y}(t) &= \mathbb{E}[e^{it(X+Y)}] \\ &= \mathbb{E}[e^{itX} e^{itY}] \\ &= \mathbb{E}[e^{itX}] \mathbb{E}[e^{itY}] \\ &= \varphi_X(t) \varphi_Y(t).\end{aligned}$$

Thus

$$\boxed{\varphi_{X+Y}(t) = \varphi_X(t) \varphi_Y(t)}$$

for independent X, Y .

More generally, for independent $X_1, \dots, X_n$,

$$\varphi_{\sum_{i=1}^n X_i}(t) = \prod_{i=1}^n \varphi_{X_i}(t).$$

This converts convolution of distributions into multiplication of functions.

Some Characteristic Functions

For

$$X \sim N(\mu, \sigma^2),$$

$$\varphi_X(t) = \exp\left(it\mu - \frac{\sigma^2 t^2}{2}\right).$$

For

$$X \sim \text{Bernoulli}(p),$$

$$\varphi_X(t) = (1 - p) + pe^{it}.$$

For

$$X \sim \text{Poisson}(\lambda),$$

$$\varphi_X(t) = \exp\{\lambda(e^{it} - 1)\}.$$

Unlike moment generating functions, characteristic functions **always exist**.

This makes them particularly useful for proving convergence in distribution.

Lévy's Continuity Theorem

Characteristic functions form another convergence-determining class.

Theorem (Lévy's Continuity Theorem)

Let φ_n be the characteristic function of X_n .

If

$$X_n \xrightarrow{d} X,$$

then

$$\varphi_n(t) \rightarrow \varphi_X(t) \quad \text{for every } t \in \mathbb{R}.$$

Conversely, if

$$\varphi_n(t) \rightarrow \varphi(t) \quad \text{for every } t,$$

and φ is continuous at $t = 0$, then φ is the characteristic function of some random variable X and

$$X_n \xrightarrow{d} X.$$

Thus, instead of working directly with CDFs, we can prove

$$\varphi_{X_n}(t) \rightarrow \varphi_X(t).$$

The Classical Central Limit Theorem

Theorem (Central Limit Theorem)

Let $X_1, X_2, \dots$ be i.i.d. with

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty.$$

Then

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1).$$

To prove this using characteristic functions, define

$$Y_i = \frac{X_i - \mu}{\sigma}.$$

Then

$$\mathbb{E}[Y_i] = 0, \quad \mathbb{E}[Y_i^2] = 1,$$

and

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i.$$

Thus it suffices to study the characteristic function of a normalized sum.

CLT via Characteristic Functions: Local Expansion

Let

$$\varphi(t) = \mathbb{E}[e^{itY_1}]$$

be the characteristic function of Y_1 .

Since

$$\mathbb{E}[Y_1] = 0, \quad \mathbb{E}[Y_1^2] = 1,$$

the characteristic function has the expansion

$$\varphi(t) = 1 - \frac{t^2}{2} + o(t^2) \quad \text{as } t \rightarrow 0.$$

Indeed,

$$e^{ity} = 1 + ity - \frac{t^2 y^2}{2} + o(t^2 y^2),$$

so taking expectations gives

$$\varphi(t) = 1 + it\mathbb{E}[Y_1] - \frac{t^2}{2}\mathbb{E}[Y_1^2] + o(t^2).$$

Hence

$$\varphi(t) = 1 - \frac{t^2}{2} + o(t^2).$$

The first two moments determine the second-order behavior of the characteristic function near the origin.

CLT via Characteristic Functions

Let

$$S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i.$$

By independence,

$$\varphi_{S_n}(t) = \left[\varphi\left(\frac{t}{\sqrt{n}}\right) \right]^n.$$

Using

$$\varphi(u) = 1 - \frac{u^2}{2} + o(u^2),$$

we obtain

$$\varphi\left(\frac{t}{\sqrt{n}}\right) = 1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right).$$

CLT via Characteristic function

Therefore,

$$\varphi_{S_n}(t) = \left[1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right) \right]^n.$$

Since

$$\left(1 + \frac{a_n}{n} \right)^n \rightarrow e^a \quad \text{if } a_n \rightarrow a,$$

we conclude that

$$\boxed{\varphi_{S_n}(t) \rightarrow e^{-t^2/2}.$$

But $e^{-t^2/2}$ is the characteristic function of $N(0, 1)$.

Hence, by Lévy's continuity theorem,

$$\boxed{S_n \xrightarrow{d} N(0, 1).$$

Why Characteristic Functions Are Powerful

The CLT proof illustrates the general strategy:

distribution of a sum $\longrightarrow$ product of characteristic functions.

For independent variables,

$$\varphi_{\sum_i x_i}(t) = \prod_i \varphi_{x_i}(t).$$

For normalized sums, we evaluate each characteristic function **near zero**.

The local expansion

$$\varphi_X(t) = 1 + it \mathbb{E}[X] - \frac{t^2}{2} \mathbb{E}[X^2] + o(t^2)$$

explains why the first two moments govern the Gaussian limit.

Remark:

The Gaussian distribution appears because

$$e^{-t^2/2}$$

is the limit of the product of many characteristic functions evaluated at increasingly small arguments.

Weak Convergence and Almost Sure Representations

Convergence in distribution is weaker than almost sure convergence:

$$X_n \xrightarrow{a.s.} X \implies X_n \xrightarrow{d} X,$$

but the reverse implication is false.

However, there is a remarkable representation result.

Theorem (Skorokhod Representation)

Suppose

$$X_n \xrightarrow{d} X.$$

Then there exist random variables

$$\tilde{X}_n, \tilde{X}$$

defined on a common probability space such that

$$\tilde{X}_n \stackrel{d}{=} X_n, \quad \tilde{X} \stackrel{d}{=} X,$$

and

$$\boxed{\tilde{X}_n \xrightarrow{a.s.} \tilde{X}.}$$

What Skorokhod Representation Does Not Say

Skorokhod representation does **not** imply

$$X_n \xrightarrow{a.s.} X$$

on the original probability space.

Instead, it constructs a new coupling

$$(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X})$$

with the correct marginal distributions:

$$\tilde{X}_n \stackrel{d}{=} X_n, \quad \tilde{X} \stackrel{d}{=} X.$$

The joint dependence structure may be completely different.

For example, recall

$$X \sim N(0, 1), \quad X_n = -X.$$

Then

$$X_n \stackrel{d}{=} X,$$

so

$$X_n \xrightarrow{d} X,$$

although X_n does not converge to X almost surely.

Skorokhod says that there exists *another coupling* with these same marginals for which almost sure convergence holds.

Why Almost Sure Representations Are Useful

Suppose

$$X_n \xrightarrow{d} X.$$

Skorokhod allows us to work with versions satisfying

$$\tilde{X}_n \xrightarrow{a.s.} \tilde{X}.$$

This lets us apply familiar pointwise arguments.

For example, if g is continuous,

$$g(\tilde{X}_n) \xrightarrow{a.s.} g(\tilde{X}).$$

Therefore,

$$g(\tilde{X}_n) \xrightarrow{d} g(\tilde{X}),$$

and since the marginal distributions are preserved,

$$g(X_n) \xrightarrow{d} g(X).$$

Thus Skorokhod provides another way to understand the Continuous Mapping Theorem:

weak convergence can often be analyzed through pathwise convergence of suitable copies.

Three Views of the Sample Mean

Let $X_1, X_2, \dots$ be i.i.d. with

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty,$$

and define

$$S_n = \sum_{i=1}^n (X_i - \mu).$$

The Law of Large Numbers tells us

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} 0.$$

The Central Limit Theorem tells us

$$\frac{S_n}{\sigma\sqrt{n}} \xrightarrow{d} N(0, 1).$$

These answer different questions:

LLN: Where does the sample mean converge?

CLT: What is its distribution at scale $n^{-1/2}$?

But neither describes the largest fluctuations that occur *along a single sample path*.

The Law of the Iterated Logarithm

The CLT describes the distribution of S_n at a fixed large n .

The LIL describes the long-run extremes of the entire sequence

$$S_1, S_2, S_3, \dots$$

Theorem (Law of the Iterated Logarithm)

Let $X_1, X_2, \dots$ be i.i.d. with

$$\mathbb{E}[X_i] = \mu, \quad 0 < \sigma^2 = \text{Var}(X_i) < \infty.$$

Then, almost surely,

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sigma \sqrt{2n \log \log n}} = 1$$

and

$$\liminf_{n \rightarrow \infty} \frac{S_n}{\sigma \sqrt{2n \log \log n}} = -1.$$

Thus the pathwise fluctuations of S_n have envelope

$$\sqrt{2n \log \log n}.$$

CLT versus LIL

At first glance, the CLT suggests

$$S_n = O_P(\sqrt{n}).$$

Indeed,

$$\frac{S_n}{\sqrt{n}} \xrightarrow{d} N(0, \sigma^2).$$

But the LIL says that almost surely,

$$S_n$$

reaches values approximately as large as

$$\pm \sigma \sqrt{2n \log \log n}$$

infinitely often.

There is no contradiction.

- $O_P(\sqrt{n})$ concerns the distribution at a **fixed** large n .
- The LIL concerns extreme fluctuations over the **entire infinite sequence**.

Thus

$\text{typical fluctuation at time } n \sim \sqrt{n},$

whereas

$\text{eventual pathwise envelope} \sim \sqrt{n \log \log n}.$